Process Characterization Plan

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCP-004

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Approvals . . . 3
Abbreviations . . . 3

1 Purpose and scope **5**

2 Objectives **6**

3 Related documents **7**

4 Prior knowledge and quality risk basis **9**
4.1 Unit-operation description and prior knowledge . . . 9
4.2 Quality attributes in scope . . . 10
4.3 Risk-based prioritization of parameters . . . 11

5 Materials and methods **12**
5.1 Scale-down model and its qualification . . . 12
5.2 Operation . . . 12
5.3 Analytical methods . . . 13
5.4 Sampling plan . . . 14
5.5 Statistical methods . . . 15

6 Study design **16**
6.1 Factors, ranges and study type . . . 16
6.2 Univariate assessment . . . 16

7 Acceptance and decision criteria **18**

8 Proven acceptable ranges (planned analysis) **19**

9 Data management and integrity **20**

10 Roles and responsibilities **21**

11 Deliverables and schedule **22**

12 Risks and assumptions **23**

13 Approvals **24**

Appendix A. Planned assessment levels **25**

Appendix B. Planned sampling and test schedule **26**

Field	Value
Document ID	PCP-004
Document class	Process Characterization Plan
Title	Process Characterization Plan: Harvest and Clarification (Step 4)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate and complete; AMV, analytical method validation; BLA, Biologics License Application; CQA, critical quality attribute; DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; LIMS, laboratory information management system; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PD, Process Development; QA, Quality Assurance; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal field; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure.

1 Purpose and scope

This plan defines the process characterization study for Harvest and Clarification, Step 4 of the A-Mab drug substance process. It states what will be studied, how the study will be run and the criteria by which its results will be judged. It is issued before any characterization run is executed.

Harvest separates the antibody from the cells and the cell debris that produced it. At commercial scale the operation is a continuous disk-stack centrifugation followed by depth filtration and a sterilizing grade filter, and it is run once per batch on the contents of a 15,000 L production bioreactor. The step forms no quality attribute of the antibody. What it decides is the condition of the feed delivered to Protein A capture, which is how much intact cell material and fine debris the clarified harvest carries and how much host cell protein and DNA is released by cells that break during the operation rather than in the reactor.

In scope are the three parameters listed in §6.1, assessed on the qualified scale-down model of §5.1 over ranges wider than the normal operating range of each. The responses are the impurity burden of the clarified harvest, its particulate content, the size-variant profile of the antibody across the step and the step yield. The study supports the transfer of the drug substance process from Novacyte Biologics — Cambridge, MA (Development) to Novacyte Biologics — Grafton, WI (Commercial DS), and it forms part of the Stage 1 process design record for the commercial process (U.S. Food and Drug Administration 2011).

Four things sit outside this plan. The culture conditions that set the impurity burden entering harvest are characterized under PCP-003 and are not varied here. The parameters of the capture step are characterized under PCP-005 and are not varied here either, and capture is used in this study only as a fixed test of the material harvest delivers (§6.2). The number and the arrangement of the depth filter stages are fixed by SOP-2006 and are not treated as study parameters. No viral clearance is claimed for this step, and no viral spiking study is included (International Council for Harmonisation 2023).

2 Objectives

The study has five objectives.

1. Qualify a scale-down model of the commercial harvest train and show that it represents the centrifugation and depth filtration performance of that train (§5.1).
2. Establish the effect of the centrifugal field, over 6,000 g to 12,000 g, on the clarity of the centrate and on the host cell protein and DNA released by cell breakage.
3. Establish the effect of depth filter load, over 80 L/m2 to 160 L/m2, on the particulate content of the clarified harvest and on the pressure the filter train develops.
4. Establish whether the step alters the size-variant profile of the antibody, and establish the effect of post-clarification turbidity, over 1 NTU to 20 NTU, on the material presented to capture.
5. Define a proven acceptable range for each parameter, confirm that the normal operating range lies inside it, and provide the evidence on which each parameter will be classified under SOP-4001.

Classification is an outcome of the study and is reported in PCR-004. No parameter is classified in this plan, and no result is stated in it.

3 Related documents

The documents that scope this study and the documents it feeds are listed in Table 3.1. RA-001 decides which parameters are studied and how (§4.3). PCR-004 reports what this plan executes.

Table 3.1: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-004	Process Characterization Report (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and the validated methods that govern the study are listed in Table 3.2. SOP-1001 governs the qualification of the scale-down model, SOP-2005 and SOP-2006 the two stages of the operation, and SOP-4001 the classification that follows from the results.

Table 3.2: Controlled procedures and validated methods for the harvest study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

At commercial scale the culture is fed continuously to a disk-stack centrifuge. A cell settles towards the bowl wall at a velocity proportional to the centrifugal field and to the difference between its own density and the density of the medium, so raising the field carries every cell and every fragment of debris outwards faster. Raising the feed rate shortens the time an element of fluid spends in the bowl in inverse proportion, and a particle that has not reached the wall when the fluid leaves is swept out in the centrate. The centrate therefore leaves the bowl still carrying the fine debris and the colloidal material that is too small to sediment in the time available. That material is removed on a depth filter train, whose media are positively charged and therefore adsorb DNA fragments and fine debris in addition to retaining particles by size. The filtrate passes a sterilizing grade filter into the harvest vessel and is held there until capture.

Two mechanisms limit the operation, and they act in opposite directions. Shear in the feed zone and at the disk stack rises with the speed of the bowl, and cells damaged by that shear release host cell protein and DNA into the centrate. A higher field therefore clarifies more completely and lyses more. The setting chosen for the step is a compromise between the two, and the purpose of the centrifugation series in §6.2 is to find where each end of that compromise begins to matter.

A depth filter loses removal capacity as the solids it has already retained fill the void space of its matrix and occupy its charged sites. As the media saturate, the pressure drop rises, more fine material passes, and at the extreme cells and debris are forced through the pores. How far the media have saturated depends on the mass of solids each unit of media area has received, so a batch volume that exhausts a small filter train leaves a large one barely loaded. The parameter is therefore carried as load per unit area, in L/m2.

Post-clarification turbidity is a measure of the fine particulate and colloidal material the filters passed. It is not a set parameter. It is a property of the clarified harvest that the centrifugal field and the filter load produce together. It is monitored on every batch as an in-process control and is listed among the parameters of the step in Table 6.1, because the value it reaches bounds what capture receives. The material it measures fouls the Protein A column and shortens the working life of the resin.

Harvest by continuous centrifugation followed by depth filtration is the platform recovery operation at Novacyte Biologics. It has been used for three previously licensed antibodies (X-Mab, Y-Mab and Z-Mab), and the equipment, the filter media grade and the control scheme are unchanged for A-Mab. Prior knowledge therefore establishes the mechanisms described above and the approximate position of the operating point. Host cell protein and DNA enter harvest at a concentration that rises as the viability of the culture falls, because cells that died before the harvest began have already released their contents into the medium. Viability at the end of culture is a property of the A-Mab culture and is characterized under PCP-003, and the ranges claimed here will therefore be established on A-Mab material.

4.2 Quality attributes in scope

Three quality attributes are in scope, and they are listed in Table 4.1 with the criticality assigned to each and the acceptance criterion that applies to the drug substance.

Table 4.1: Quality attributes in scope at harvest, with the drug substance acceptance criteria.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60

The Tool #1 column is the attribute score returned by the ranking tool of the case study this process model follows (CMC Biotech Working Group 2009). It is the impact of the attribute multiplied by the uncertainty in that impact, and the criticality grade beside it is assigned separately. Aggregate carries the highest score of the three and residual DNA the lowest.

None of the three is formed at this step. Host cell protein and DNA enter harvest in the culture fluid and leave it in the clarified harvest, and the step neither clears them nor is credited with clearing them. Aggregate is formed upstream and is reduced principally at cation exchange. All three are nonetheless in scope, because the operation can add to the first two by lysis and, in principle, to the third by shear at the feed zone or at an air-liquid interface in a foaming transfer.

The acceptance criteria in Table 4.1 apply to the drug substance. They are met by the clearance the purification train delivers downstream of this step, and they are not in-process limits for the clarified harvest. No in-process limit is derived for harvest from them, and none is set in this plan. The criteria this step will be judged against are stated in §7, and with one exception they are comparative.

Host cell protein and aggregate carry most of the weight in Table 4.1. Host cell protein is the response through which the lysis argument of §4.1 is tested, and it is the attribute that rises most when cells break during the operation. Aggregate carries a higher criticality than host cell protein but a much lower expectation of an effect, since the exposure to shear at this step is brief. Residual DNA is of very low criticality as a drug substance attribute, and it is measured here because DNA fragments and nucleosomes foul depth filters and Protein A resin, which makes it informative about the condition of the feed even where its own criterion is far from binding.

4.3 Risk-based prioritization of parameters

The parameters brought into this study, and the study type assigned to each, follow the pre-characterization risk assessment RA-001. That assessment ranked each parameter of the step on its potential to affect a quality attribute and on its potential to interact with another parameter, and it assigned a study type on the combined ranking. All three parameters of this step were assigned univariate assessment, which is recorded in the study type column of Table 6.1. No parameter of this step was ranked high enough to require a multivariate design.

The step forms no quality attribute, so a parameter of the step can reach one only by breaking cells, which raises host cell protein and DNA in the clarified harvest, or by unfolding the antibody at a shear or air-liquid interface, which raises high molecular weight content. Both mechanisms act in one direction only, and neither can lower an attribute the step does not form. In addition, the impurities released are cleared by three downstream steps, so the consequence of a poorly chosen setting is a heavier load on capture and not a change in the drug substance.

The range to be studied is wider than the normal operating range on both sides of every parameter (Table 6.1). A range that stops at the normal operating limits cannot show where the operation begins to fail, and an excursion beyond a normal operating limit is then a deviation with no data behind it. The ranges also cover the settings a receiving site might need if its equipment differs from the development one, which is part of what the transfer requires of this study (International Council for Harmonisation 2008).

5 Materials and methods

5.1 Scale-down model and its qualification

The characterization runs will be executed on a scale-down model of the harvest train, qualified under SOP-1001 before the first characterization run.

The model reproduces the commercial operation in three parts, and each part is matched on the quantity that drives its own mechanism and not on the size of the equipment. Matching on volume or on bowl diameter alone would leave the shear at the feed zone, the settling in the bowl and the loading of the filter media free to differ between the scales. Cell damage in the feed zone will be reproduced in a bench shear device operated at the energy dissipation rate calculated for the commercial feed zone. Sedimentation will be reproduced in a laboratory centrifuge operated at the same relative centrifugal field as the commercial bowl and at a matched ratio of flow to sigma factor, so that a setting expressed in g means the same thing at both scales. Depth filtration will be reproduced with disks of the commercial filter media, run at the same flux and to the same load per unit area, so that a setting expressed in L/m2 carries across in the same way.

Qualification will compare the model with the commercial train at the set-point condition on the same culture. The comparison will cover centrate turbidity, solids removal, host cell protein and DNA in the clarified harvest, filter pressure at the end of the load, and step yield. Replicate runs will be performed at each scale. The acceptance criterion is that no performance measure differs between the scales at $\alpha = 0.05$. Where a difference is found, its cause should be identified and the model should be modified before the characterization runs begin.

A commercial disk-stack centrifuge discharges solids periodically and recirculates part of its contents, and the number of passes an element of fluid makes through the feed zone is not reproduced at bench scale. The model therefore represents the field and the residence time well and the cumulative shear history less well, and it is expected to under-represent lysis rather than to over-represent it. This study uses the model to bound ranges and to establish the direction and the approximate size of each effect. It does not use the model to predict the absolute impurity concentration of a commercial batch.

5.2 Operation

Each characterization run will be executed as a single pass of the whole harvest through the model train, in the order the commercial operation runs it.

The culture will be taken from the production bioreactor scale-down model at the end of culture, at the conditions characterized under PCP-003. Culture from the qualified bioreactor model matters here, because a culture that ends at lower viability enters harvest with more host cell protein already released into the medium and with more fragile cells, and both of those raise the impurity concentration of the clarified harvest whatever the centrifuge is doing. Where the volume allows, one culture will be split between the levels of a parameter, so that the levels of a series are compared on the same material.

The culture will be fed to the centrifuge model at the field setting for the run and at the matched ratio of flow to sigma factor. The centrate will be collected, sampled, and passed over the depth filter train at constant flux to the load per unit area for the run, and then over a sterilizing grade filter into the harvest container. Temperature will be held at the commercial set-point throughout. The time between the end of culture and the end of clarification will be held constant across the runs, since a longer hold on unclarified culture allows further lysis and would confound the comparison the study is making.

Every parameter not varied in a run will be held at its set-point, and the set-points are those given in Table 6.1. Runs will be executed against the procedures listed in Table 3.2.

5.3 Analytical methods

Four validated methods report the responses of this study, and they are listed with the procedures in Table 3.2.

Turbidity by nephelometry (AMV-3015) measures the fine particulate and colloidal material the filters pass, and it is the in-process measurement by which clarified harvest is released to capture. Host cell protein by ELISA (AMV-3012) measures the impurity burden of the clarified harvest, and it is the response that carries the lysis argument of §4.1. Residual DNA by qPCR (AMV-3014) measures the second product of lysis and the material the depth filter media adsorb. Size variants by SEC-HPLC (AMV-3011) measure the high molecular weight content of the antibody before and after the step.

The validated performance of the two methods whose figures bound the decisions of §7 is given in Table 5.1.

Table 5.1: Validated performance of the harvest impurity and turbidity methods. AMV-3011 and AMV-3014 are validated under their own reports and are not quoted here.

Reference	Method	Precision (%)	LOQ	LOQ unit	Accuracy (%)	Measurement variance (fraction of total)
AMV-3015	Turbidity by Nephelometry (NTU)	4	0.5	NTU	98	0.38
AMV-3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95	0.4

The host cell protein ELISA is the less precise of the two, at 9.5 % with a limit of quantitation of 1 ng/mg, and 40.0% of the variance of a single result is attributable to the method. A change in host cell protein across a parameter range that is smaller than the method can resolve will therefore not be reported as an effect, and §5.5 states how that comparison will be made. Turbidity is measured more precisely, at 4 % with a limit of quantitation of 0.5 NTU. Precision matters more for turbidity than the figure alone suggests, because the turbidity criterion in §7 is an absolute limit and not a comparison against a reference sample.

Samples for host cell protein and for DNA will be assayed in one analytical campaign per parameter series, with the same reagent lots where the schedule allows. A trend across the levels of a parameter should not be confounded by a change of reagent lot part way through the series.

5.4 Sampling plan

Samples will be drawn at four points in every run. The points, the methods applied at each and the purpose of each are listed in Appendix B.

The sample of unclarified culture is the reference for the whole study. Impurity released by the operation can only be seen as a difference from what entered it, so that sample will be drawn from the production bioreactor scale-down model at the end of culture, immediately before the feed to the centrifuge starts, and it will be assayed in the same campaign as the samples that follow it. The centrate sample separates the two stages of the operation. It shows what the centrifuge passed and therefore what the depth filter was asked to remove, which is the quantity the bracketing arrangement of §5.5 varies. The sample after depth filtration and the sample after sterilizing filtration are both clarified harvest, and the second of them is the material capture receives.

Each level will be sampled once at each point. The set-point condition will be run in replicate across the execution of the study, and those replicates provide the estimate of run to run variability against which every effect is tested. Samples for size variants will be drawn at the first and the last point only, since the step is not expected to change the size-variant profile and the intermediate points would not localize a change if one were found.

5.5 Statistical methods

Each response will be analysed against the parameter that was varied, by linear regression on the levels run, with the replicates at the set-point condition providing an independent estimate of error. Effects will be tested at $\alpha = 0.05$. Where the relationship is not linear over the range, a quadratic term may be fitted, and the fit will be reported with the terms that were needed rather than with the terms that were planned.

Statistical significance and practical significance will be reported separately. A significant slope will be reported together with the change it predicts across the normal operating range of that parameter, and that change will be compared with the precision of the method that measured the response (§5.3). An effect smaller than the method can resolve will be reported as detected and not material, with the comparison shown.

No adequacy threshold is set for the regressions. The decision in §7 rests on the size of an effect against the precision of its method, not on the fit of a predictive model, so each regression will be reported with its coefficient of determination and its residual standard error and will be judged on those.

Results below the limit of quantitation of a method will be reported as such and excluded from the regression, and the number excluded will be given with the fit. Residual DNA is the response most likely to be affected, because the depth filter media adsorb DNA and the assay then reports a small quantity.

The design is one factor at a time. It cannot determine an interaction between two parameters, where the effect of one parameter on a response depends on the level of the other. One interaction is expected on mechanistic grounds, since a higher centrifugal field passes fewer solids to the depth filter and so changes the load that filter has to handle. That interaction will be bracketed and not estimated. The depth filter load series will be run on centrate produced at 6,000 g and again on centrate produced at 12,000 g, so that the filter is challenged with both extremes of the solids load it can receive. The bracket bounds the filter response between the two solids loads the centrifuge can deliver, and it yields no coefficient for the interaction, so none will be reported.

No response-surface model will be fitted for this step and no Monte-Carlo propagation of a fitted model will be performed, because no multivariate design is run here. A range reported from this study therefore holds with the other parameters at their set-points, and it says nothing about two of them moving away from their set-points together.

6 Study design

6.1 Factors, ranges and study type

The parameters, their set-points, the ranges to be studied, the normal operating ranges and the study type assigned to each are given in Table 6.1.

Table 6.1: Parameters in scope, ranges to be studied and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Centrifugation (rcf)	g	9,000	6000–12000	8000–10000	univariate
Depth filter load	L/m2	120	80–160	100–140	univariate
Post-clarification turbidity	NTU	5	1–20	1–10	univariate

Three rows carry the design, and they do not carry it in the same way. Centrifugation is the parameter with two opposed effects, and its range runs from 6,000 g to 12,000 g so that both ends of the compromise described in §4.1 are reached. Depth filter load runs to 160 L/m2, which loads the media well beyond the upper normal operating limit of 140 L/m2 and past the point at which the pressure drop is expected to rise. Post-clarification turbidity is the response of the other two parameters (§4.1), and the range in its row is the range over which turbidity will be presented to capture, not a range of settings that will be dialled in at the centrifuge.

The set-point column is the condition at which every parameter not being varied will be held, and it is the condition at which the replicate runs of §5.4 will be executed. The study type column records the outcome of the risk assessment described in §4.3.

6.2 Univariate assessment

Each parameter will be assessed at five levels. They are the lower edge of the range studied, the lower limit of the normal operating range, the set-point, the upper limit of the normal operating range and the upper edge of the range studied, and they are listed in Appendix A. Three of the five are limits that already exist in the control strategy, so the study reports directly on the limits it is meant to justify.

Runs will be executed in randomized order within each parameter series. A drift in the culture, in the equipment or in an assay over the execution of a series would otherwise be read as an effect of the parameter. Replicates at the set-point condition will be distributed across a series rather than run together, for the same reason.

The centrifugation series will be run first, because its outcome supplies the two centrate qualities used to bracket the depth filter load series (§5.5). The depth filter load series will follow. The turbidity series will be run last, because the material for it is produced by the two series before it. Clarified harvest at a target turbidity will be prepared by blending clarified harvest with retained centrate to the target turbidity for the level, which allows the turbidity presented to capture to be varied independently of the settings that produced it. Blended material carries the particle size distribution of the centrate it was made from, and that is not identical to the distribution a depth filter at the end of its capacity would pass.

The turbidity series will be loaded onto a scale-down Protein A column held at the set-point conditions of PCP-005. The responses are the pressure developed across the column over the load and the yield of the cycle. No conclusion about the capture step will be drawn from these runs. Capture is used here as a fixed test of the material harvest delivers, and the characterization of capture itself is reported in PCR-005.

7 Acceptance and decision criteria

This step forms no quality attribute, and no in-process limit is derived for it from the drug substance criteria (§4.2). The criteria below are therefore comparative, with one exception. They are fixed here so that the analysis of §8 applies a rule agreed before execution rather than a judgement formed once the data are visible.

Impurity release is the first criterion. Host cell protein and residual DNA in the clarified harvest will be compared with the same measurements on the unclarified culture that produced it. A level passes when no increase is detected at $\alpha = 0.05$, or when a detected increase is smaller than the resolution of the method that measured it (§5.3). A setting that releases more impurity than that lies outside the acceptable range of its parameter.

Clarification is the second criterion and the absolute one. Post-clarification turbidity of the clarified harvest will be compared with the upper limit of its normal operating range, 10 NTU. A centrifuge setting or a filter load that delivers clarified harvest above that value has failed to clarify, whatever the impurity results show.

Product quality is the third. High molecular weight content will be compared before and after the step, and a level passes when no increase is detected at $\alpha = 0.05$. A null result is required here rather than a bounded one, because aggregate formed at harvest would be carried into capture and would consume clearance that the cation exchange step is credited with delivering.

Step yield is reported but is not gated. Yield will be given for every level and compared with the set-point replicates. No numeric yield limit is set in this plan, because none follows from a product quality requirement at this step. A loss of yield at a level will be reported with its cause and will be weighed in the choice of the operating range. It will not on its own place a level outside the proven acceptable range.

A parameter meets the criteria over a range when every level in that range passes the first three criteria. The proven acceptable range of the parameter is then that range, bounded at the levels tested (§8). Where a level fails, the proven acceptable range is bounded at the last level that passed, and the failure should be investigated before the range is reported.

A failure at a level inside the normal operating range means the operating range in force is not supported by the data, and it should be raised to Manufacturing Science and to Quality Assurance before the study continues, so that the normal operating range or the control strategy can be changed under SOP-4001. Deviations from this plan during execution will be recorded, investigated and reported in PCR-004 with their impact on the conclusions, whether or not the affected runs are retained.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a parameter over which the step has been shown to meet its criteria, with every other parameter held at its set-point (International Council for Harmonisation 2009). A range established by varying one parameter at a time holds while the other parameters are at their set-points (§5.5), and no combination of two parameters away from their set-points is claimed for this step. The bracketing arrangement of the depth filter load series is the only place where the study looks at two parameters together, and it bounds the response between two solids loads rather than modelling it.

A proven acceptable range is also bounded by the levels tested. The analysis will not extrapolate beyond the edges given in Table 6.1. Where every level of a parameter meets the criteria, the proven acceptable range will be reported as the whole range studied and not as anything wider.

For each parameter, PCR-004 will give the range studied, the criterion applied, the response that came closest to that criterion, and the proven acceptable range that follows. The normal operating range will be shown nested inside it. Where the two coincide at an edge, that will be stated, because a normal operating limit sitting on the edge of the proven range leaves no margin for the ordinary variability of the commercial operation.

No design space is claimed for this step. A design space is a multivariate region, and this study runs no multivariate design. What Step 4 contributes to the control strategy is a proven acceptable range for each parameter, a normal operating range nested inside it, and an in-process control on the turbidity of the clarified harvest. The classification of each parameter under SOP-4001 will be made in PCR-004 on the evidence this study produces, and it will roll up into PCMR-001 with the other steps of the train.

9 Data management and integrity

Data from this study will be recorded, reviewed and retained under the site data governance procedures, and the ALCOA+ principles apply to every record the study produces.

Runs will be executed against a pre-approved batch record for the scale-down train. Each entry will be made at the time it is observed and signed by the person who made it. Analytical results will be acquired and stored in the validated laboratory information management system, with the audit trail enabled and reviewed as part of result approval. Equipment used to set or to measure a parameter of this study will be within its calibration at the time of use, and the calibration status will be recorded with the run.

Raw data, the analysis script and the outputs of that script will be retained together, so that the tables in PCR-004 can be regenerated from the raw data without a manual step. Any change to a recorded value will be made as a corrected entry, with the reason and the original value visible. The analysis will be verified by a second person in Statistics, and the report will be reviewed by the functions listed in Table 1.

10 Roles and responsibilities

Responsibility for the study rests with Process Development. The functions that must agree with it, and what each is accountable for, are listed in Table 10.1.

Table 10.1: Functions and responsibilities for the harvest characterization study.

Function	Responsibility
Process Development / MSAT	Owns this plan, executes the runs on the scale-down model, interprets the results and writes PCR-004
Analytical Development	Releases the methods, performs the assays and reports method performance with the results
Manufacturing Science (receiving site)	Confirms the commercial operation the scale-down model represents and reviews the ranges proposed
Statistics	Agrees the analysis before execution, performs it, and verifies the analysis of another
Quality Assurance	Approves this plan and PCR-004, and approves the disposition of any deviation

Two of these functions are engaged before execution rather than after it. The analysis of §5.5 will be agreed with Statistics before the first run, so that the model to be fitted is not chosen once the data are visible. The commercial operation the scale-down model represents will be confirmed with Manufacturing Science at the receiving site before qualification, since the model is matched to that operation and not to the development one (International Council for Harmonisation 2008).

11 Deliverables and schedule

The deliverable of this study is the Process Characterization Report for Harvest and Clarification, PCR-004. It will report the qualification of the scale-down model, the results of each parameter series, the proven acceptable ranges, the classification of each parameter and any deviation from this plan.

The study will run in four phases, in order. The scale-down model will be qualified first, and no characterization run will be executed until that qualification has been reviewed and accepted. The centrifugation series will follow, then the depth filter load series on centrate from the extremes of the centrifugation range, then the turbidity series on material prepared from both. Analysis of a series may begin once the last analytical result of that series has been approved, and the report will be issued when all three series have been analysed.

No dates are fixed in this plan. The schedule is held in the transfer plan PTP-001, where it is coordinated with the other characterization studies of the campaign and with the availability of the bioreactor scale-down model.

12 Risks and assumptions

Five things could weaken the conclusions of this study, and each carries a mitigation or a bound.

The scale-down model is the largest of them. A continuous disk-stack centrifuge is the hardest unit in this process to represent at bench scale, for the reason given in §5.1, and the model is expected to under-represent cumulative shear. The mitigation is that the study reports effects and ranges rather than absolute impurity levels, and that the qualification of §5.1 compares the model with the commercial train on the responses that matter here.

The study assumes the culture is representative. A culture that ends at lower viability carries more debris and more broken cells into harvest, which raises both the solids load on the centrifuge and the host cell protein released before the run begins. Culture will therefore be taken from the qualified bioreactor scale-down model at its set-point condition, and the viability and the cell density of every culture used will be recorded with the run and reported in PCR-004.

Material supply constrains what can be compared within a series. One culture may not provide enough volume for all five levels of a parameter at the scale of the model, so a series may span more than one culture. Where it does, the set-point replicates run on each culture provide the reference for the levels run on that culture, and the analysis will fit the culture as a block.

The blended material of the turbidity series carries the particle size distribution of centrate and not that of a saturated depth filter (§6.2). The turbidity range will therefore be reported as a range of turbidity delivered to capture, and no claim will be made about the specific material a filter at the end of its capacity would pass.

Assay variability bounds the smallest effect the study can see. The host cell protein ELISA is the less precise of the two methods quoted in Table 5.1, and the first criterion of §7 is written against its precision rather than against a fixed percentage. Residual DNA falls close to the limit of quantitation of the qPCR assay once the depth filter media have adsorbed part of it, and a result below that limit carries no magnitude to regress (§5.5).

13 Approvals

This plan takes effect when it has been approved by every function listed in Table 1. Execution will not begin before that approval. Any change to the plan after approval will be made as a revision to this document and approved by the same functions. A change made during execution that is not a revision to this plan is a deviation, and it is handled as §7 describes.

Appendix A. Planned assessment levels

Every level in Table 13.1 is a value that already exists in the control strategy for this step. Three of the five are limits the study is meant to justify, and the other two are the edges of the range studied.

Table 13.1: Planned assessment levels for each parameter, in the units of Table 6.1.

Parameter	Unit	Lower range edge	Lower NOR limit	Set-point	Upper NOR limit	Upper range edge
Centrifugation (rcf)	g	6,000	8,000	9,000	10,000	12,000
Depth filter load	L/m2	80	100	120	140	160
Post-clarification turbidity	NTU	1	1	5	10	20

Appendix B. Planned sampling and test schedule

Table 13.2 lists the four sample points of every run, the methods applied at each and what each point is for. The methods are those of Table 3.2.

Table 13.2: Planned sampling points and the methods applied at each.

Sample point	Methods	Purpose
Unclarified culture, drawn before the feed to the centrifuge starts	AMV-3015, AMV-3012, AMV-3014, AMV-3011	Reference against which impurity release and any size-variant change are judged
Centrate, after centrifugation	AMV-3015, AMV-3012, AMV-3014	What the centrifuge passed, and the solids load the depth filter receives
Filtrate, after depth filtration	AMV-3015, AMV-3012, AMV-3014	What the depth filter media retained and adsorbed
Clarified harvest, after sterilizing filtration	AMV-3015, AMV-3012, AMV-3014, AMV-3011	The material presented to Protein A capture

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2023. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.